BASERA

Student Accommodation Discovery & Comparison Platform

FRONT END WEB DEVELOPMENT
26CA205PCB

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SYNOPSIS PRESENTATION

A student-first platform for finding the right place to stay.

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Project Overview

What Basera is, why it is needed, and how it creates value.

The problem

- Students relocating for college often depend on brokers, social-media groups and scattered listings.
- Rent, deposit, distance, room type and amenities are difficult to compare consistently.
- Students may have limited knowledge of nearby localities and suitable price ranges.
- Repeated calls and manual shortlisting make the decision process time-consuming.

Problem statement

A centralized, student-oriented accommodation discovery and comparison experience is needed.

The Basera solution

01	**Search**	College, locality, area or property
02	**Filter**	Budget, type, room, rating & amenities
03	**Evaluate**	Structured details, distance & reviews
04	**Shortlist**	Favourites and side-by-side comparison
05	**Connect**	Enquiry flow for selected properties

Semester 1: React.js • JavaScript • HTML5 • CSS3
Semester 2: MERN full-stack application

ONE PLACE
Centralized discovery

COMPARE
Student-first evaluation

DECIDE
More informed shortlisting

Motivation, Significance & Key Findings

MOTIVATION

Why this project matters:

01

Student-first approach

Focuses on college proximity, budget, sharing, food, Wi-Fi, AC, security and reviews.

02

Real-world relevance

Accommodation is a recurring challenge for students moving to a new city.

03

Value to both sides

Students reduce fragmented searching; owners get a structured channel to reach them.

04

Academic progression

A practical frontend prototype now, with a clear path to MERN, APIs and persistent data.

KEY FINDINGS FROM ANALYSIS

The real challenge is not merely displaying listings — it is helping students evaluate them consistently.

Important design constraint

Claims like “verified” or “available now” require reliable data sources and validation. Semester 1 therefore uses sample data; real verification is deferred to the backend/admin phase.

DESIGN PRINCIPLE • SEARCH → FILTER → COMPARE → ENQUIRE

Project Objectives & Success Criteria

What Basera aims to achieve .

OBJECTIVES

1. **Discover** — Search by college, locality, area or property.
2. **Filter** — Budget, type, room, rating and amenities.
3. **Understand** — Present property information in a consistent format.
4. **Shortlist** — Save favourites and revisit preferred properties.
5. **Compare** — Evaluate selected properties side by side.
6. **Connect** — Provide enquiry and owner-listing flows.
7. **Scale** — Keep the frontend ready for MERN integration.

SUCCESS CRITERIA

✓	**Responsive**	Works on desktop, tablet and mobile layouts.
✓	**Functional search**	Search and filters return the expected prototype results.
✓	**End-to-end journey**	Student can discover → evaluate → shortlist → enquire.
✓	**shortlist**	Favourites and comparison flows operate consistently.
✓	**MERN-ready**	Frontend structure supports future APIs, auth and persistence.

Semester 1 success = a working frontend not a production database system.

Project Scope, Modules & Boundaries

SCOPE

What is included in Basera — and what is intentionally outside the initial prototype.

STUDENT / USER

- **Search & filters**
- **Property details**
- **Favourites**
- **Comparison**
- **Enquiry**

PROPERTY OWNER

- **Submit listing**
- **Manage listing***
- **View enquiries***
- **Profile status***

* *Full-stack phase*

ADMINISTRATOR

- **Manage users***
- **Moderate listings***
- **Handle reports***
- **Verification/status***

* *Full-stack phase*

OUTSIDE INITIAL SCOPE

Online rent/payment collection • Legal tenancy contracts • Guaranteed real-time availability • Automatic owner verification • Native mobile app

Core product idea: PG / flat / hostel discovery for students — not a generic real - estate marketplace.

Analysis Phase & System Flow

ANALYSIS

User roles, requirements and the primary student journey.

FUNCTIONAL REQUIREMENTS

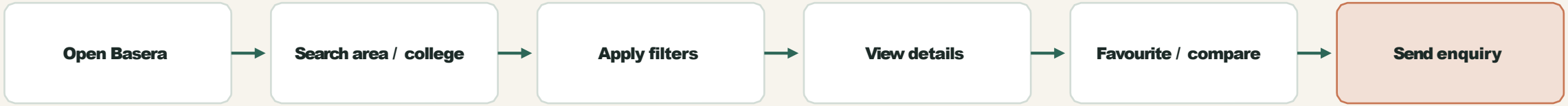
- Search by college, locality, area or property name.
- Display PG / flat / hostel / room listings with filters.
- Provide detailed property views and amenity information.
- Add/remove favourites and compare properties side by side.
- Provide an enquiry action for a selected property.
- Support owner listing submission.

NON-FUNCTIONAL REQUIREMENTS

Usability	Clear navigation for first-time users.
Responsiveness	Desktop, tablet and mobile support.
Performance & reliability	Efficient filter response; graceful errors.
Security*	Authentication, authorization and validation.

** Backend phase*

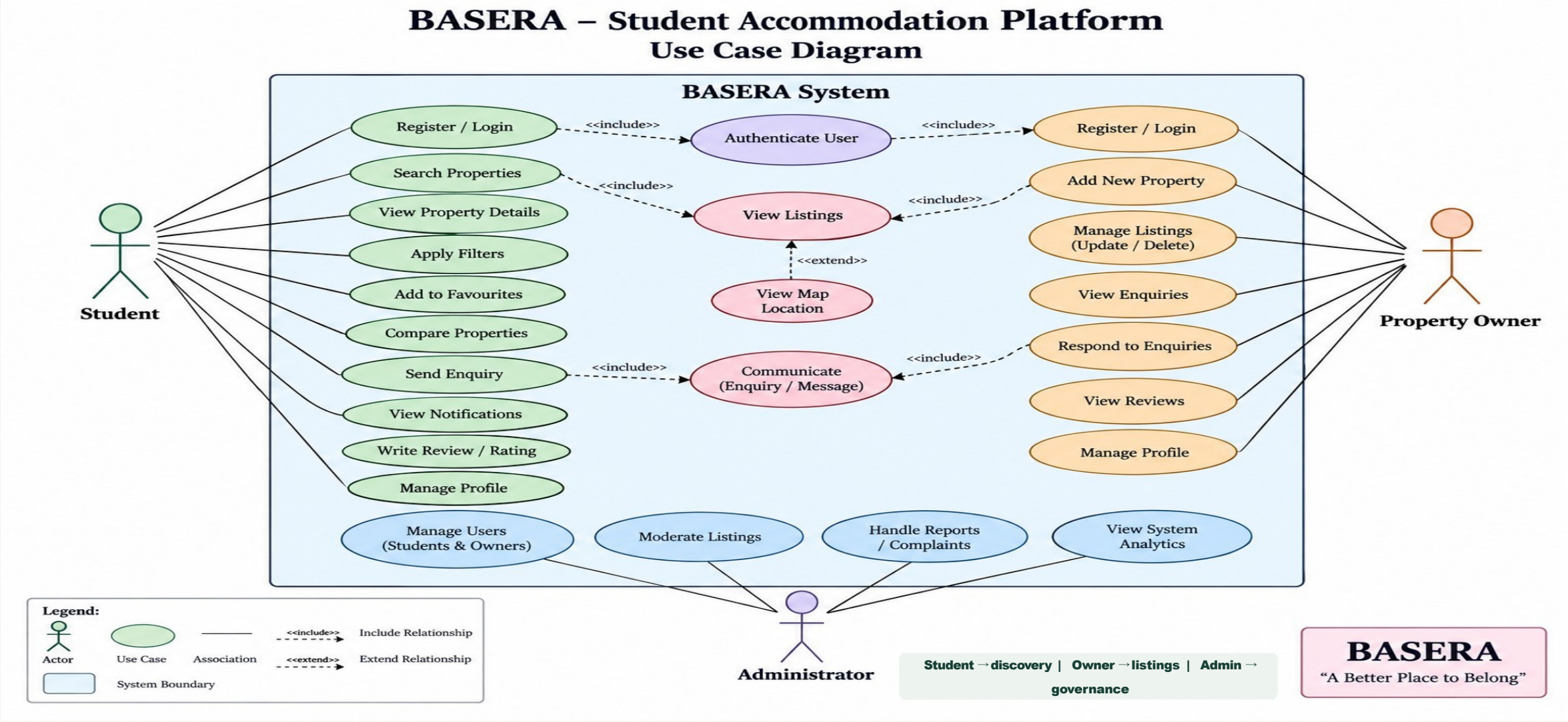
PRIMARY SYSTEM FLOW



Use Case Diagram

Actors and the major interactions planned for the Basera system.

SYSTEM ANALYSIS



Data Flow Diagram — Level 0

DFD LEVEL 0

Context diagram: Basera as one system interacting with three external entities.

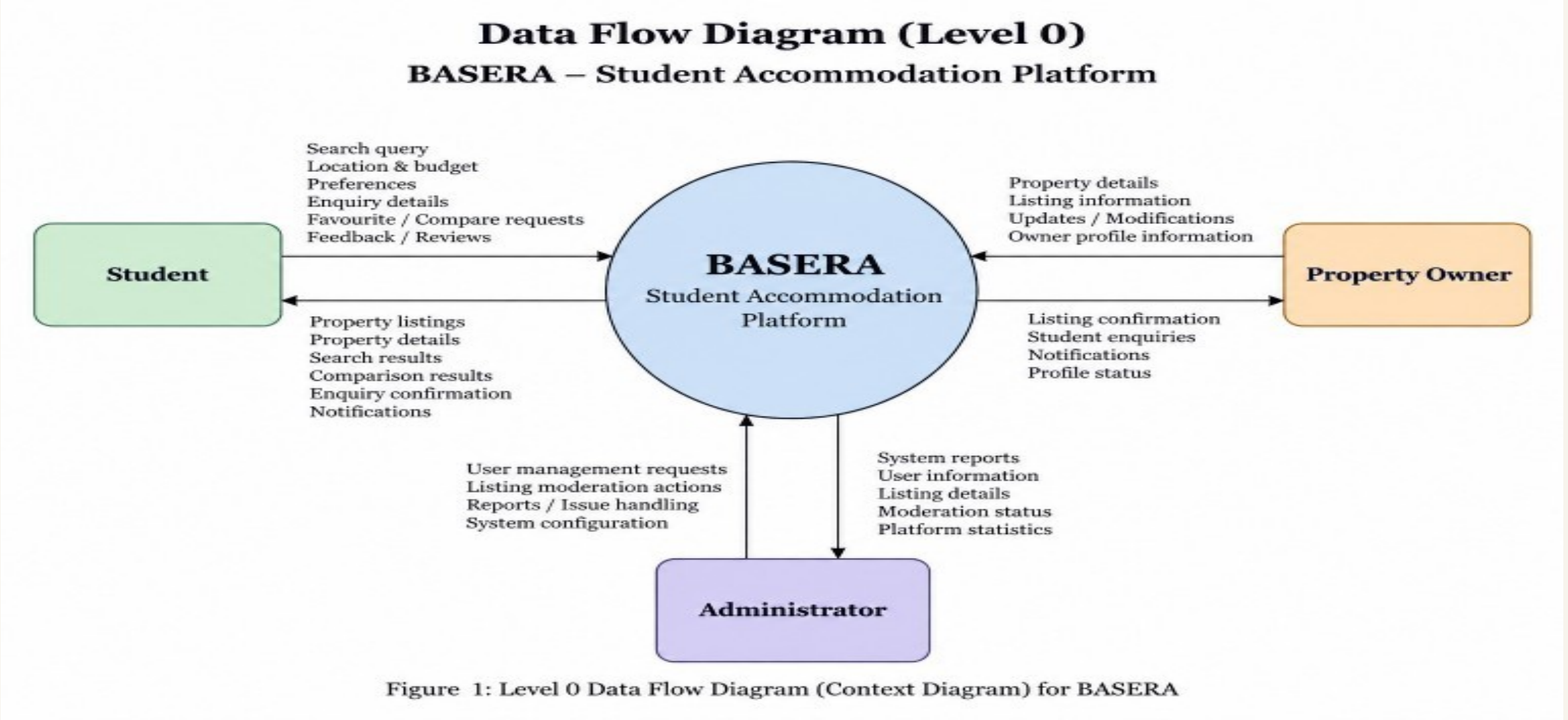


Figure 1: Level 0 Data Flow Diagram (Context Diagram) for BASERA

Data Flow Diagram — Level 1

How the Basera system decomposes into major processes and data stores.

DFD LEVEL 1

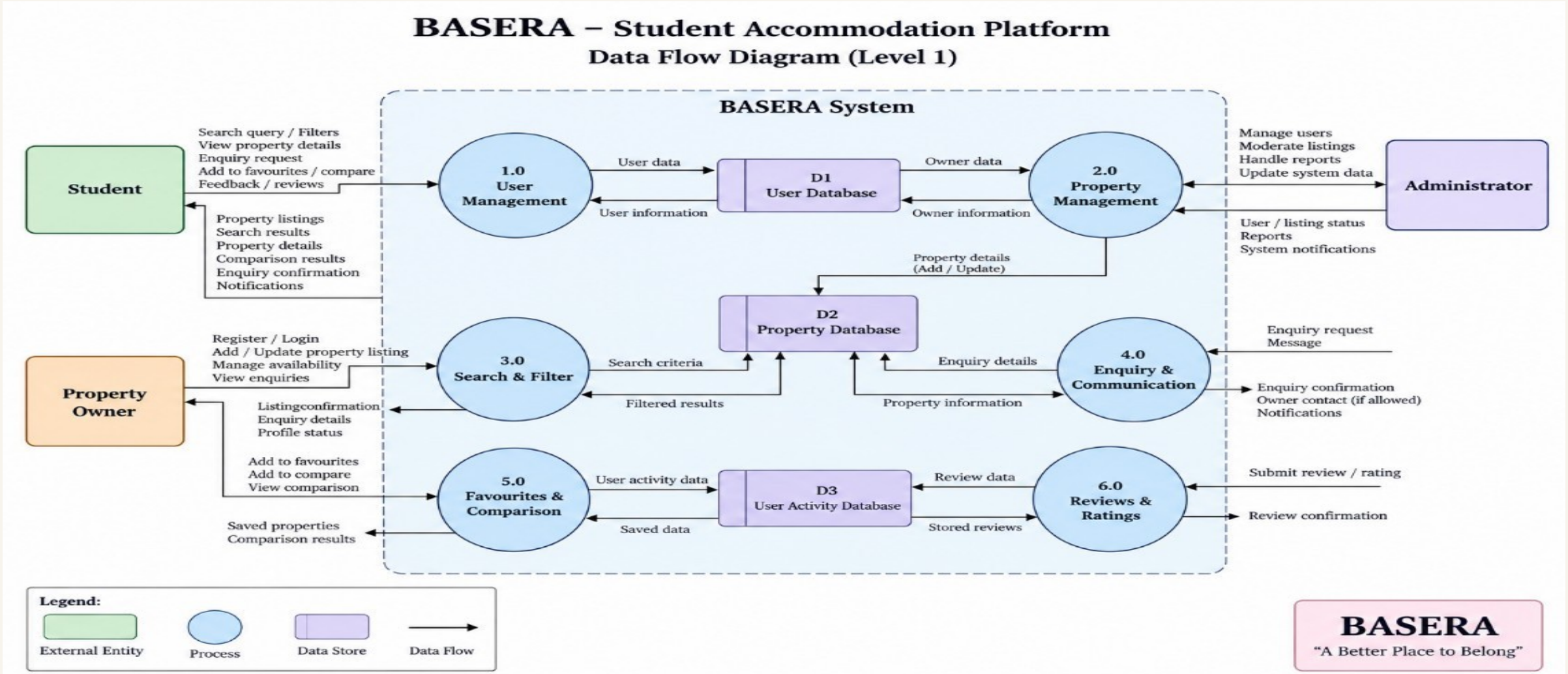


Figure 2: Level 1 Data Flow Diagram for BASERA

Technology Stack, Architecture & Requirements

Semester 1 frontend, Semester 2 MERN architecture

in the final phase.

TECHNOLOGY

SEMESTER 1 • FRONTEND

React.js • JavaScript • HTML5 • CSS3

Responsive static pages, property listing layouts, navigation, forms, and sample accommodation data.

SEMESTER 2 • MERN

Node + Express • MongoDB

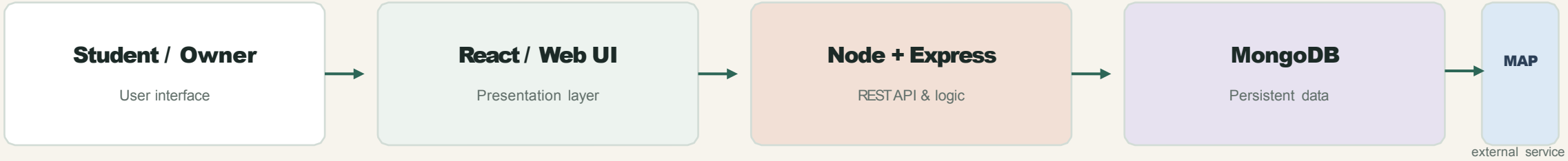
REST APIs, authentication, validation, role-based logic and persistent data.

TOOLS & SERVICES

Figma • VS Code • Git/GitHub Postman • Maps API • Deployment

Design, development, version control, API testing, location support and hosting.

SYSTEM ARCHITECTURE



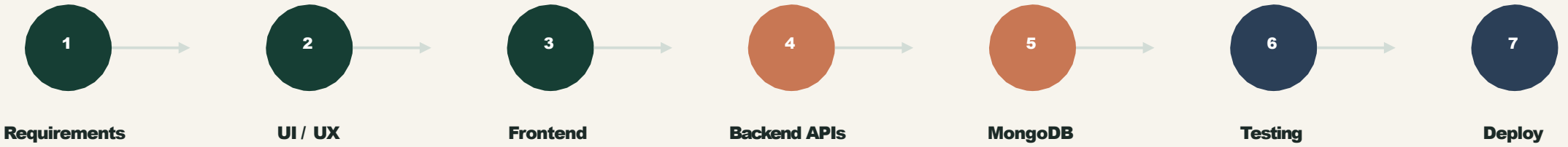
HARDWARE	Core i3 / equivalent+ • 8 GB RAM recommended • Stable internet	SOFTWARE	Modern browser • VS Code • Node.js • MongoDB • Git

Methodology & Project Timeline

METHODOLOGY

Incremental Agile-style development across the two-semester journey.

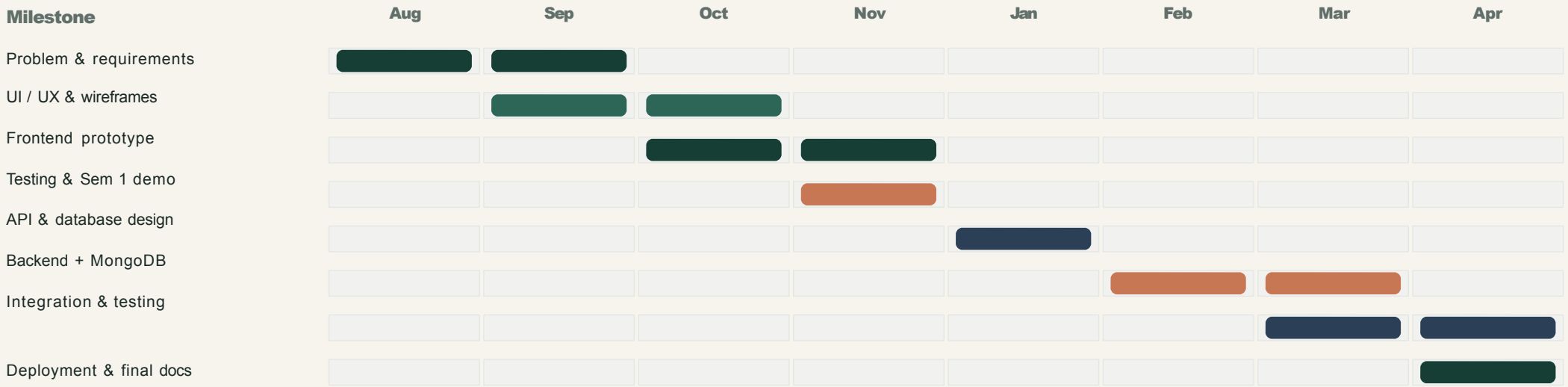
AGILE-STYLE DEVELOPMENT



SEMESTER 1 • Prototype, user journey, responsive frontend

SEMESTER 2 • MERN backend, persistence, roles, integration & deployment

PROPOSED GANTT TIMELINE



Expected Outcomes & Future Direction

OUTCOME

What this project delivers now — and where Basera can grow next.

PRIMARY PROJECT OUTCOME • WEB APPLICATION DEVELOPMENT

SEMESTER 1 — PROTOTYPE

- ✓ Responsive multi-device frontend using React.js • JavaScript • HTML5 • CSS3
- ✓ Static search & filter
- ✓ Detailed property views
- ✓ favourites & comparison page
- ✓ Owner submission + enquiry UI options

SEMESTER 2 — FULL STACK

- ✓ Node/Express + MongoDB
- ✓ Authentication & role-based access
- ✓ Persistent properties, reviews & enquiries
- ✓ Owner/admin management workflows
- ✓ End-to-end deployment

FUTURE SCOPE

- ✓ AI-based accommodation recommendations
- ✓ Roommate matching
- ✓ Map search & travel distance
- ✓ Property / owner verification
- ✓ Rental insights, alerts & mobile app

REFERENCES :

supporting links & references

MDN Web Docs • React • Node.js • Express.js • MongoDB • Git/GitHub • Figma

Useful Links:

- ✓ https://developer.mozilla.org/en-US/
- ✓ https://react.dev/learn
- ✓ https://nodejs.org/docs/latest/api/
- ✓ https://expressjs.com/
- ✓ https://www.mongodb.com/docs/

TEAM Member's Github Accounts:

- ✓ https://github.com/Shivani-Warrior/BaseraFinal
- ✓ https://github.com/Shanu-lab-bot/Basera.git
- ✓ https://github.com/2628mca0156-creator/Basera.git

THANK YOU